

KAVI MOOKHERJEE AMODT

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EDUCATION

UC Berkeley

Expected August 2026

B.A. Computer Science and Cognitive Science; Linguistics Minor

GPA: 3.72

Relevant Coursework: Probability & Random Processes, Artificial Intelligence, Algorithms, Computer Security, Software Engineering, Algorithmic Economics, Computability & Complexity, Econometrics, Computer Architecture, Data Structures, Discrete Math & Probability, Linear Algebra, Statistics

PROJECTS

Secure File Sharing System

- Designed and implemented a cryptographically secure file sharing system in Go using public-key encryption, digital signatures, and HMACs.
- Validated correctness under a fully adversarial server environment where attackers hold full write access to the datastore and read access to the keystore.

Regression and Gradient Descent From Scratch

- Implemented linear and logistic regression from first principles with gradient descent and closed-form solvers in Python/NumPy.
- Added ridge (L2) and lasso (L1) regularization and tuned penalty strength via k-fold cross-validation, using bias-variance analysis to diagnose over- and underfitting.

Assembly Level Digit Classifier

- Implemented a neural network inference engine in RISC-V assembly, capable of classifying handwritten digits through matrix operations and ReLU activation.

MCMC Cipher Decoder

- Implemented the Metropolis-Hastings algorithm from scratch in Python/NumPy to sample from arbitrary target distributions.
- Applied MCMC to decrypt substitution ciphers by modeling key permutations as a Markov chain.

Prediction Market Simulator

- Simulated and visualized prediction market dynamics using the LMSR online learning algorithm in C++.

EXPERIENCE

Software Engineering Intern

May 2024 – Aug. 2024

Tools for Humanity

San Francisco, CA

- Developed production Rust, Python, and Bash code for an iris-scanning biometric device (the Orb).
- Engineered Rust CLI tooling to surface and query real-time Orb device metrics.
- Gained exposure to Docker and Kubernetes for containerized deployments, and wrote scripts to automate testing workflows.

Research Assistant

Jan. 2025 – Present

Language and Cognitive Development Lab

UC Berkeley

- Designed and implemented computational experiment for word-object learning models using dynamic field theory in MATLAB.
- Developed automated data processing and analysis pipelines in Python.

Research Assistant

2021 – 2023

UCSC Linguistics Department

Santa Cruz, CA

- Conducted eye-tracking research on syntactic ambiguity resolution and lexical underspecification.

SKILLS

Languages: Python, Go, C++, Ruby, SQL, C, Java, R, RISC-V Assembly, Bash

Libraries & Frameworks: PyTorch, NumPy, SciPy, pandas, Matplotlib; Rails

Tools & Platforms: Git, Docker, Kubernetes, Jupyter, Linux, MATLAB, Model Context Protocol (MCP)

Domains: Machine learning, deep learning, applied cryptography, information theory, probabilistic modeling